day_5

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```
#RNA-Seq Gene Expression Normalization Using RPKM and TPM and Heatmap
Visualization
counts <- read.table("raw_count.xlsx", header = TRUE, sep = "\t", stri
ngsAsFactors = FALSE)
```

```
anno <- read.table("Scer_annotation.txt", header = FALSE, sep = "\t",
stringsAsFactors = FALSE)
head(anno)
```

```
##            V1   V2    V3    V4 V5        V6
## 1 NC_001133.9 gene  1807  2169  -   YAL068C
## 2 NC_001133.9 gene  2480  2707  + YAL067W-A
## 3 NC_001133.9 gene  7235  9016  -   YAL067C
## 4 NC_001133.9 gene 11565 11951  -   YAL065C
## 5 NC_001133.9 gene 12046 12426  + YAL064W-B
## 6 NC_001133.9 gene 13363 13743  - YAL064C-A
```

```
colnames(anno) <- c("Chr", "Feature", "Start", "End", "Strand", "Gen
e")
anno$Length <- anno$End - anno$Start + 1
gene_length <- anno[, c("Gene", "Length")]
head(anno)
```

```
##           Chr Feature Start   End Strand      Gene Length
## 1 NC_001133.9    gene  1807  2169      -   YAL068C    363
## 2 NC_001133.9    gene  2480  2707      + YAL067W-A    228
## 3 NC_001133.9    gene  7235  9016      -   YAL067C   1782
## 4 NC_001133.9    gene 11565 11951      -   YAL065C    387
## 5 NC_001133.9    gene 12046 12426      + YAL064W-B    381
## 6 NC_001133.9    gene 13363 13743      - YAL064C-A    381
```

```
data <- merge(counts, gene_length, by.x = "Genes", by.y = "Gene")
head(data)
```

```
##   Genes ConA_R1 ConA_R2 ConB_R1 ConB_R2 Length
## 1 Q0045       3       5      21      19  12884
## 2 Q0050       5       1      25      32   2505
## 3 Q0055       4       7      33      34   5013
## 4 Q0060       0       1       3       3   6179
## 5 Q0065       0       0       0       2   8118
## 6 Q0070       1       0       0       0   9350
```

```
library_size <- colSums(data[,2:5])
library_size
```

```
##  ConA_R1  ConA_R2  ConB_R1  ConB_R2
##  9263208  9015358 16593821 16476494
```

```
#Calculated RPKM
data$RPKM_ConA_R1 <- (data$ConA_R1 * 1e9) / (data$Length *library_size
["ConA_R1"])
data$RPKM_ConA_R2 <- (data$ConA_R2 * 1e9) / (data$Length * library_siz
e["ConA_R2"])
data$RPKM_ConB_R1 <- (data$ConB_R1 * 1e9) / (data$Length * library_siz
e["ConB_R1"])
data$RPKM_ConB_R2 <- (data$ConB_R2 * 1e9) / (data$Length * library_siz
e["ConB_R2"])
data$Length_kb <- data$Length/1000
head(data$RPKM_ConA_R1)
```

```
## [1] 0.02513675 0.21547697 0.08613921 0.00000000 0.00000000 0.011545
88
```

```
# Calculated RPK
data$RPK_ConA_R1 <- data$ConA_R1 / data$Length_kb
data$RPK_ConA_R2 <- data$ConA_R2 / data$Length_kb
data$RPK_ConB_R1 <- data$ConB_R1 / data$Length_kb
data$RPK_ConB_R2 <- data$ConB_R2 / data$Length_kb

# Calculated Sums
sumA1 <- sum(data$RPK_ConA_R1)
sumA2 <- sum(data$RPK_ConA_R2)
sumB1 <- sum(data$RPK_ConB_R1)
sumB2 <- sum(data$RPK_ConB_R2)
```

```
#Calculated TPM

data$TPM_ConA_R1 <- (data$RPK_ConA_R1 / sumA1) * 1e6
data$TPM_ConA_R2 <- (data$RPK_ConA_R2 / sumA2) * 1e6
data$TPM_ConB_R1 <- (data$RPK_ConB_R1 / sumB1) * 1e6
data$TPM_ConB_R2 <- (data$RPK_ConB_R2 / sumB2) * 1e6
```

```
#Saved the results
write.csv(data, "RPKM_TPM_results.csv", row.names=FALSE)

colSums(data[, c("RPKM_ConA_R1", "RPKM_ConA_R2", "RPKM_ConB_R1", "RPKM
_ConB_R2")])
```

```
## RPKM_ConA_R1 RPKM_ConA_R2 RPKM_ConB_R1 RPKM_ConB_R2
##     984933.1     989207.0    1013108.1    1000995.7
```

```
colSums(data[, c("TPM_ConA_R1", "TPM_ConA_R2", "TPM_ConB_R1", "TPM_Con
B_R2")])
```

```
## TPM_ConA_R1 TPM_ConA_R2 TPM_ConB_R1 TPM_ConB_R2
##       1e+06       1e+06       1e+06       1e+06
```

```
# 1. Extract TPM columns
expr <- data[, c("TPM_ConA_R1", "TPM_ConA_R2", "TPM_ConB_R1", "TPM_Con
B_R2")]

# 2. Set gene names as matrix row names
rownames(expr) <- data$Genes

# 3. Convert to numeric matrix
expr <- as.matrix(expr)
mode(expr) <- "numeric"

# 4. Log2 transformation with pseudo-count
expr_log <- log2(expr + 1)
head(expr_log)
```

```
##       TPM_ConA_R1 TPM_ConA_R2 TPM_ConB_R1 TPM_ConB_R2
## Q0045  0.03635742  0.06145274  0.13350323  0.12355241
## Q0050  0.28542970  0.06317613  0.67233437  0.82744710
## Q0055  0.12095824  0.20986231  0.47671783  0.49695256
## Q0060  0.00000000  0.02594628  0.04107507  0.04185673
## Q0065  0.00000000  0.00000000  0.00000000  0.02139118
## Q0070  0.01681364  0.00000000  0.00000000  0.00000000
```

#the heatmap

```
heatmap(expr_log, Rowv = NA, Colv = NA, scale = "none", col = heat.col
ors(100))
```

